CONCENTRATION & EXIT RISK ASSESSMENT

# AWS exit risk review — demo organisation

Synthetic data for demonstration. Not a real organisation.

| SCOPE | ORGANIZATIONAL UNIT | ACCOUNTS IN SCOPE | SPEND PERIOD |
|---|---|---|---|
| name contains 'prod' | Root | 12 of 21 | 2026-04-01 to 2026-10-01 (6 months, amortised) |
| SCANNED | MANAGEMENT ACCOUNT | RESOURCE INVENTORY | EXIT TARGET ASSUMED |
| 10/20/2026, 2:49:55 PM | 111111111111 | Included (deep scan) | Microsoft Azure |

## 1 Executive summary

**OU exit-risk score**

**43** /100

**Elevated exit risk**

Balanced (square root of spend). Other weightings: by spend 43 · per account 45




Annual spend in scope

**$7.3m**

$608k / month, amortised

Spend on High/Critical services

**32%**

$2.4m a year

Portable accounts

**7 of 12**

58% Low/Moderate · 5 High/Critical (43% of spend)

Indicative exit cost

**$7.1m**

range $5.3m – $10.6m

Indicative duration

**28 months**

range 22 – 39 months

Engineering effort

**323 person-months**

capacity 15.0 FTE

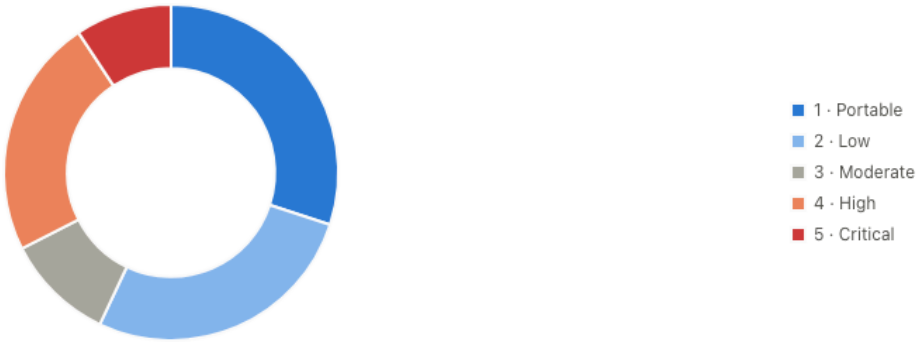
**Key messages**

- The in-scope estate scores 43/100 (Elevated) for exit difficulty, across 12 account(s) spending USD 7.3m a year.
- 32% of spend (USD 2.4m/yr) runs on High or Critical services that have no direct equivalent outside AWS and need re-engineering.
- 7 of 12 account(s) (58%) are Low or Moderate risk and broadly portable; 5 (42%) are High or Critical, holding 43% of spend.
- The largest lock-in exposures are Redshift, DynamoDB, Lambda.
- 3 account(s) - cards-processing-prod, data-platform-prod, contact-centre-prod - carry 55% of the weighted risk; they set the pace of any exit.
- A full exit to Microsoft Azure is estimated at 323 person-months over ~28 months (22-39), costing about USD 7.1m (USD 5.3m-USD 10.6m) in engineering and parallel running.

## 2 Where the exit risk sits

### Monthly spend by exit-risk tier

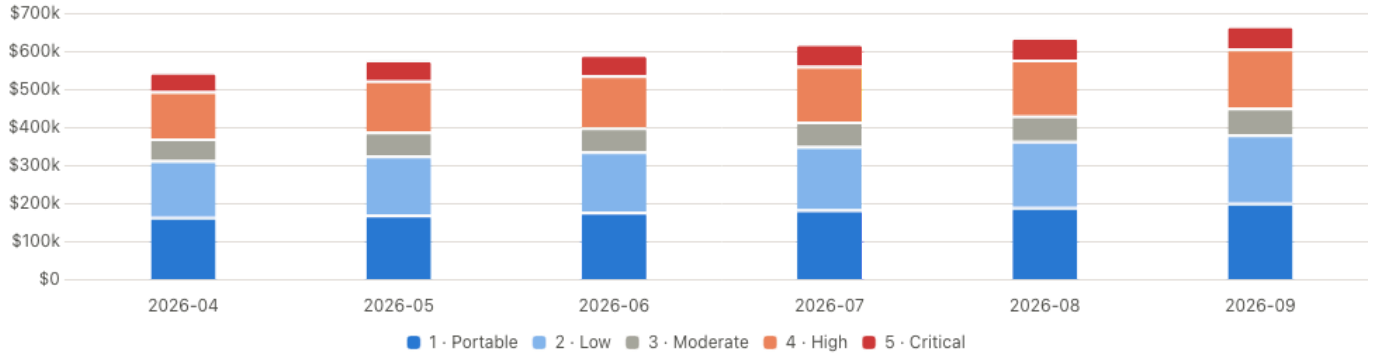
Share of in-scope spend by how hard it is to move.




| | USD / month | Share | Services | Accounts |
|---|---|---|---|---|
| 1 · Portable | $183k | 30.0% | 12 | 8 |
| 2 · Low | $164k | 27.0% | 17 | 12 |
| 3 · Moderate | $64k | 10.5% | 11 | 6 |
| 4 · High | $140k | 23.1% | 16 | 9 |
| 5 · Critical | $57k | 9.3% | 3 | 6 |

### Spend trend by tier

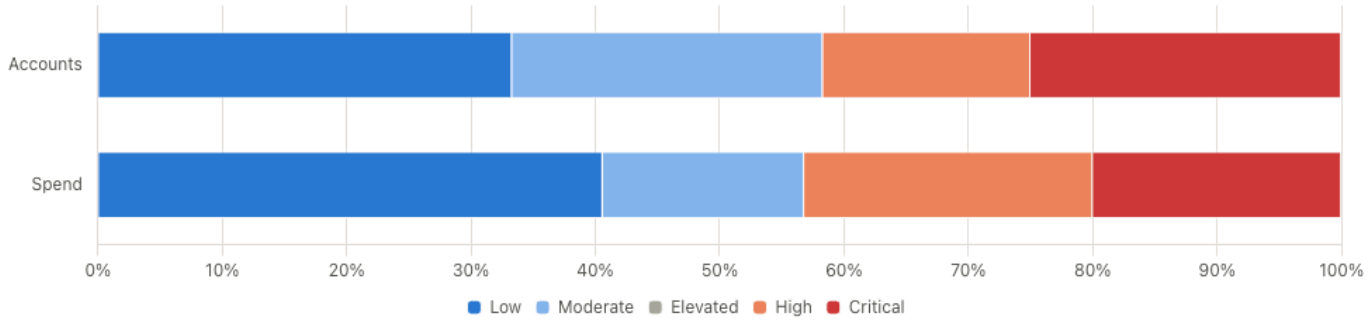
Is lock-in growing? Monthly amortised spend.



## 3 Accounts: how many are at risk, and how big

### Accounts by exit-risk rating

Share of accounts (top) against share of spend (bottom). Few accounts holding a lot of spend is a concentration to manage, not an estate-wide problem.



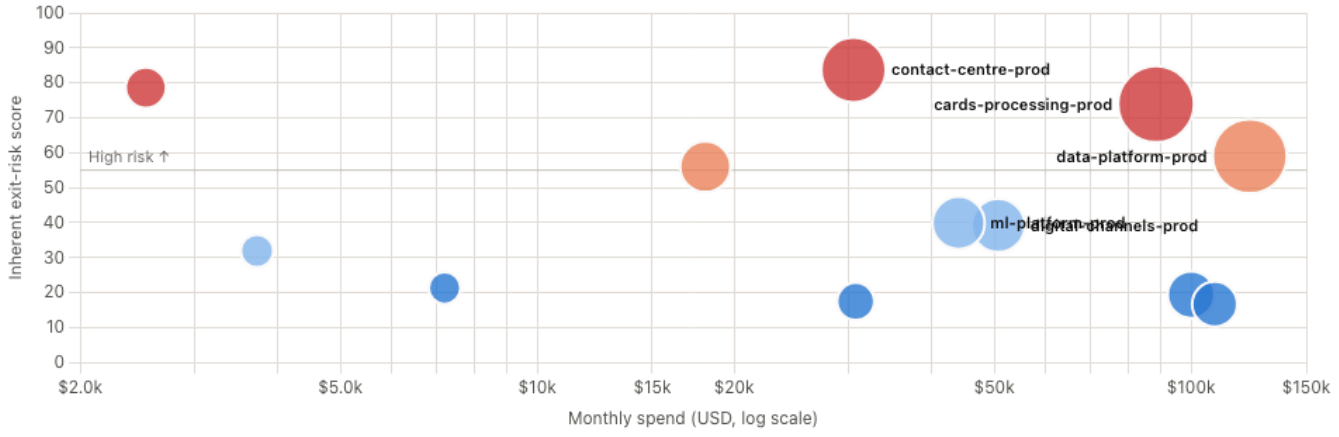

### Portable vs higher-risk accounts

| Rating | Accounts | Share | USD / month | Spend share |
|---|---|---|---|---|
| Low | 4 | 33.3% | $247k | 40.6% |
| Moderate | 3 | 25.0% | $99k | 16.2% |
| Elevated | 0 | 0.0% | $0 | 0.0% |
| High | 2 | 16.7% | $141k | 23.2% |
| Critical | 3 | 25.0% | $121k | 20.0% |
| **Portable** (Low + Moderate) | **7** | 58.3% | | |

OU score weighting: Balanced (square root of spend). Bigger accounts count for more, but one very large account cannot dominate the score. Change it in the bar at the top.

### Inherent risk vs spend

Each bubble is an account. Up = harder to move; right = bigger; bubble size = points it contributes to the OU score. Top-right accounts drive the exit.



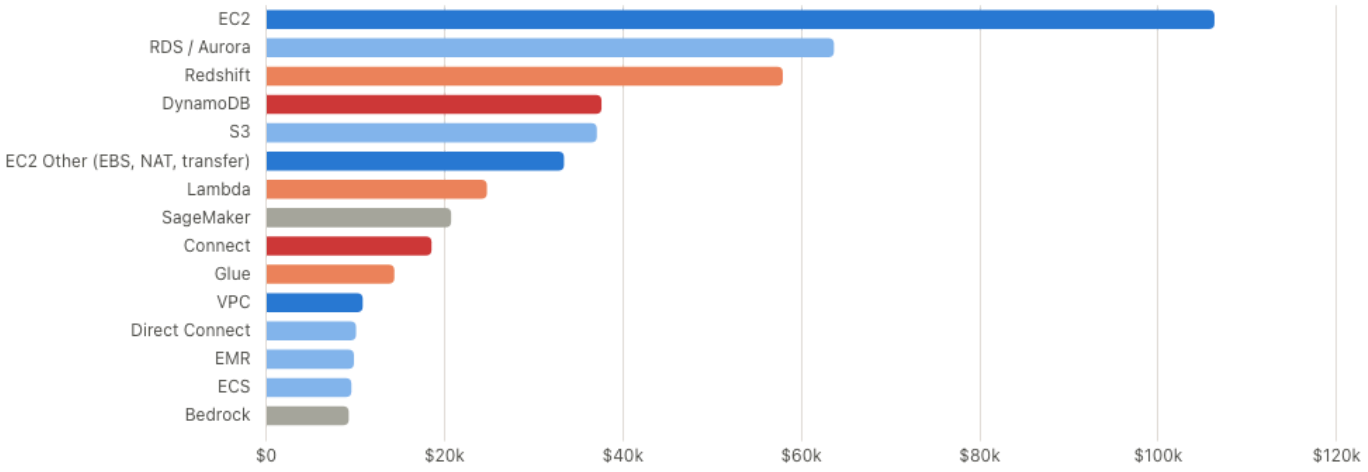
## Account ranking by weighted contribution

| # | Account | USD / month | Risk | Rating | Weight | Contribution (pts) | High/Crit % | Main lock-in | Effort (eng-wks) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **cards-processing-prod** 770736660454 · Root / Workloads / Production | $88k | 74 | Critical | 12.3% | 9.1 | 77% | DynamoDB, Lambda, Kinesis Streams | 146 |
| 2 | **data-platform-prod** 497405839480 · Root / Workloads / Production | $123k | 59 | High | 14.6% | 8.6 | 63% | Redshift, Glue, QuickSight | 157 |
| 3 | **contact-centre-prod** 834750990725 · Root / Workloads / Production | $30k | 84 | Critical | 7.2% | 6.1 | 86% | Connect, Lex, Lambda | 111 |
| 4 | **digital-channels-prod** 756384864470 · Root / Workloads / Production | $51k | 39 | Moderate | 9.3% | 3.7 | 25% | Cognito, AppSync, Pinpoint | 65 |
| 5 | **ml-platform-prod** 459490555598 · Root / Workloads / Production | $44k | 40 | Moderate | 8.7% | 3.5 | 2% | Bedrock AgentCore | 51 |
| 6 | **fraud-analytics-prod** 439439398733 · Root / Workloads / Production | $18k | 56 | High | 5.6% | 3.1 | 28% | DynamoDB, Timestream, Lambda | 58 |
| 7 | **payments-core-prod** 534439589175 · Root / Workloads / Production | $100k | 19 | Low | 13.1% | 2.5 | 4% | DynamoDB, Lambda | 64 |
| 8 | **core-banking-prod** 113442468479 · Root / Workloads / Production | $109k | 17 | Low | 13.7% | 2.3 | 0% | none | 40 |
| 9 | **treasury-reporting-prod** 890290181436 · Root / Workloads / Production | $2.5k | 79 | Critical | 2.1% | 1.6 | 92% | DynamoDB, QuickSight, Lambda | 30 |
| 10 | **shared-network-prod** 926883241220 · Root / Infrastructure | $31k | 18 | Low | 7.3% | 1.3 | 0% | none | 14 |
| 11 | **legacy-batch-prod** 800533991158 · Root / Workloads / Production | $3.7k | 32 | Moderate | 2.5% | 0.8 | 25% | SWF, Data Pipeline | 12 |
| 12 | **identity-prod** 461410438745 · Root / Infrastructure | $7.2k | 21 | Low | 3.5% | 0.8 | 0% | none | 8 |

## 4 Services driving lock-in

### Top services by spend

Coloured by exit-risk tier.



## High & critical services in use

| Service | Tier | USD / month | Share | Accounts |
|---|---|---|---|---|
| **Redshift**<br>Analytics | 4 · High | $58k | 9.5% | 1 |
| **DynamoDB**<br>Database | 5 · Critical | $38k | 6.2% | 5 |
| **Lambda**<br>Compute | 4 · High | $25k | 4.1% | 5 |
| **Connect**<br>Business apps | 5 · Critical | $19k | 3.0% | 1 |
| **Glue**<br>Analytics | 4 · High | $14k | 2.4% | 1 |
| **Kinesis Streams**<br>Analytics | 4 · High | $8.5k | 1.4% | 2 |
| **Cognito**<br>Security | 4 · High | $7.3k | 1.2% | 2 |
| **Step Functions**<br>Integration | 4 · High | $6.4k | 1.1% | 2 |
| **QuickSight**<br>Analytics | 4 · High | $5.6k | 0.9% | 2 |
| **AppSync**<br>Integration | 4 · High | $3.9k | 0.6% | 1 |
| **Lex**<br>AI/ML | 4 · High | $2.8k | 0.5% | 1 |
| **Pinpoint**<br>Business apps | 4 · High | $2.6k | 0.4% | 1 |
| **Timestream**<br>Database | 4 · High | $1.8k | 0.3% | 1 |
| **EventBridge**<br>Integration | 4 · High | $1.4k | 0.2% | 1 |
| **Amplify**<br>DevOps | 4 · High | $1.2k | 0.2% | 1 |
| **Bedrock AgentCore**<br>AI/ML | 4 · High | $1.1k | 0.2% | 1 |
| **SWF**<br>Integration | 5 · Critical | $625 | 0.1% | 1 |
| **Lake Formation**<br>Analytics | 4 · High | $604 | 0.1% | 1 |
| **Data Pipeline**<br>Analytics | 4 · High | $291 | 0.1% | 1 |

## 5 Exit cost & time estimate

### About your organisation

These answers drive the estimate. They are saved in this browser only.

Do teams use a standard CI/CD pipeline to create assets?

**Several, partially standard pipelines**

Do you use Terraform, and is it up to date?

**Terraform with significant drift**

| People in the core cloud team | % of their time on the exit |
|---|---|
| **10** | **50** |

Application engineers available (FTE)

**10**

Target platform

**Microsoft Azure**

| Blended day rate | Currency |
|---|---|
| **750** | **USD** |

| FX: currency per 1 USD | Contingency % |
|---|---|
| **1** | **25** |

| Parallel-run months | Materiality (USD/month) |
|---|---|
| **6** | **25** |

Regulated entity (exit plan evidence, regulator notification, assurance testing)

Total cost (indicative)

**$7.1m**

$5.3m engineering + $1.8m parallel run

Calendar duration

**28 months**

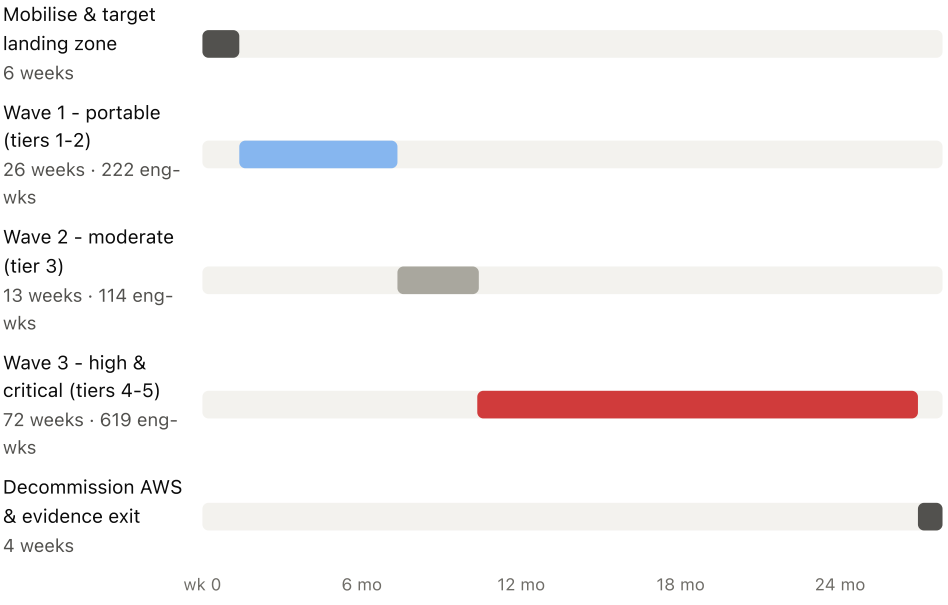
bound by team capacity

Effort

**1,400 eng-weeks**

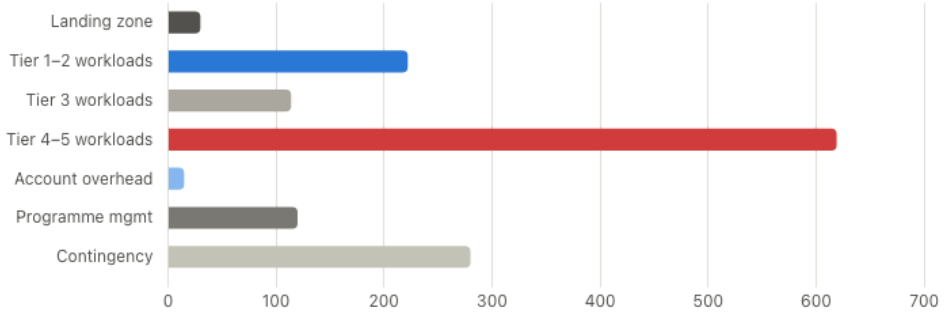
323 person-months · 12 material accounts

### Indicative timeline




### Where the effort goes

Engineer-weeks after delivery-maturity adjustment.



## Your answers and the assumptions behind the estimate

| | | |
|---|---|---|
| Standard CI/CD pipeline<br>Several, partially standard pipelines | Terraform / IaC<br>Terraform with significant drift | Core cloud team<br>10 people at 50% |
| Application engineers<br>10 FTE | Target<br>Microsoft Azure | Blended day rate<br>750 USD |
| Contingency<br>25% | Parallel running<br>6 months | Regulated<br>Yes |

- Service effort: catalog engineer-weeks per workload x sqrt(monthly spend / 1,000 USD), clamped 0.25-8.0x; resource counts from the deep scan raise it where higher.
- Services under 25 USD/month are scored but not estimated (immaterial).
- 1 engineer-week per material account for network, IAM and decommissioning.
- Target landing zone: 24 weeks (4 + 12 x sqrt(spend / 100,000 USD), max 24) + 0.5 per material account.
- Delivery factor 1.26 = CI/CD 1.0 x IaC 1.1 x target 1.0 x regulated 1.15.
- Programme management and assurance 12%; contingency 25%.
- Engineers are 80% productive on migration work; duration is the longer of effort / capacity and the critical path of the hardest tier present, plus decommissioning.
- Parallel running: 6 months at 50% of current AWS spend (target ramps up as AWS ramps down).
- Excluded: target-cloud run cost after exit, licence changes, early-termination of AWS commitments (EDP, Savings Plans, RIs), and data-transfer-out fees (AWS waives egress for customers leaving AWS on request to AWS Support, since March 2024).
- Range shown is -25% / +50% on cost and -20% / +40% on duration (early-stage estimation cone).

## 6 Data quality & scan issues

What this report does *not* cover. Every API error during the scan is recorded here and in the log.

Critical: 0 | Error: 0 | Warning: 2 | Info: 0

| Severity | Area | Account | Issue |
|---|---|---|---|
| Warning | inventory<br>ap-southeast-2 | data-platform-prod | eventbridge_rules in ap-southeast-2: Rate exceeded `ThrottlingException`<br>→ AWS throttled the request after retries. Re-run, or lower EXITRISK_MAX_WORKERS. |
| Warning | catalog | – | 'AWS Transform' is not in the service catalog; scored as Moderate (tier 3). 437/month in scope.<br>→ Add it to src/exitrisk/data/service_catalog.yaml (see CONTRIBUTING.md). |

## A Appendix: service exit-risk catalog

Every AWS service is classified into one of five tiers by how much of a workload must change to leave AWS. The tier sets the service's risk score; the recommended path and typical duration are for one workload of that service.

| Tier | Score | Meaning |
|---|---|---|
| 1 · Portable | 10 | Commodity infrastructure or an open standard. Lift-and-shift; config moves, code does not change. |
| 2 · Low | 30 | Managed flavour of an open engine or de-facto standard API. Data and config migrate; little code change. |
| 3 · Moderate | 50 | AWS-specific API with a close equivalent on every major cloud. Code changes at the integration points. |
| 4 · High | 75 | Proprietary programming or data model. The workload must be redesigned, not just re-pointed. |
| 5 · Critical | 95 | No equivalent outside AWS. Rebuild, re-platform or replace with a product; often a business change. |

### Tier 5 · Critical 3 service(s)

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **DynamoDB**<br>Database<br>$38k/mo in scope | Proprietary key-value/document model. Single-table designs, GSIs, conditional writes, TTL and DynamoDB Streams are baked into application code and data layout. No wire-compatible equivalent outside AWS. | Redesign the data access layer. Common targets are Cosmos DB, Firestore/Bigtable, ScyllaDB (Alternator offers a DynamoDB-compatible API) or PostgreSQL JSONB. Replace Streams consumers. Run dual-writes during cutover. | **Azure:** Cosmos DB<br>**GCP:** Firestore / Bigtable / Spanner<br>**Neutral:** ScyllaDB Alternator / Cassandra / PostgreSQL | 2-6 months per application |
| **Connect**<br>Business apps<br>$19k/mo in scope | Full contact-centre platform - contact flows, phone numbers, Lex bots, Lambda integrations, recordings and agent desktop. Number porting and agent retraining make it a business transformation. | Select a CCaaS platform (Genesys, NICE, Five9, Dynamics 365 Contact Center), rebuild flows, port numbers, migrate recordings under retention rules, retrain agents. | **Azure:** Dynamics 365 Contact Center<br>**GCP:** Contact Center AI Platform<br>**Neutral:** Genesys Cloud / NICE CXone / Five9 | 6-12 months |
| **SWF**<br>Integration<br>$625/mo in scope | Legacy proprietary workflow engine with a bespoke decider/worker model; no equivalent. | Rewrite on Temporal (the conceptual successor) or Durable Functions. | **Azure:** Durable Functions<br>**GCP:** Workflows<br>**Neutral:** Temporal | 2-4 months |

### Tier 4 · High 16 service(s)

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **Redshift**<br>Analytics<br>$58k/mo in scope | PostgreSQL-derived but proprietary MPP warehouse - distribution/sort keys, Spectrum external tables, stored procedures, UNLOAD/COPY from S3, Redshift ML and data sharing are all specific. Downstream BI and ELT jobs depend on its dialect. | Choose a warehouse (Snowflake or Databricks are cloud-neutral; BigQuery / Fabric Warehouse are native). Convert DDL and SQL with a translation tool, re-point ELT (dbt makes this far cheaper), reconcile results in parallel for several reporting cycles. | **Azure:** Fabric Warehouse / Synapse / Snowflake<br>**GCP:** BigQuery<br>**Neutral:** Snowflake / Databricks / ClickHouse | 3-9 months per warehouse |
| **Lambda**<br>Compute<br>$25k/mo in scope | The function code is portable, but the event model, triggers (S3, SQS, DynamoDB streams, EventBridge), IAM wiring, layers and runtime limits are AWS-specific. Real Lambda estates are mostly glue between other AWS services. | Inventory triggers first. Repackage as containers on Cloud Run / Container Apps / Knative, or rewrite as Azure Functions; replace each event source with the target's equivalent. Migrate with the services it glues together, not alone. | **Azure:** Azure Functions / Container Apps<br>**GCP:** Cloud Run functions<br>**Neutral:** Knative / OpenFaaS on Kubernetes | 4-12 weeks per application |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **Glue**<br>Analytics<br>$14k/mo in scope | Spark under the hood, but jobs use GlueContext/DynamicFrame APIs, the Glue Data Catalog, crawlers and Glue workflows. The catalog is usually shared by Athena, EMR, Redshift Spectrum and Lake Formation, so it is a hub. | Port jobs to plain Spark (Databricks, Dataproc, Synapse Spark), migrate the catalog to Unity / Hive Metastore / Dataplex, and replace crawlers with explicit schemas. | **Azure:** Data Factory + Databricks<br>**GCP:** Dataproc / Dataflow + Dataplex<br>**Neutral:** Spark + Airflow + Hive Metastore | 1-4 months |
| **Kinesis Streams**<br>Analytics<br>$8.5k/mo in scope | Proprietary streaming API (shards, KCL checkpointing in DynamoDB, enhanced fan-out). Producers and consumers must be rewritten. | Move to Kafka (MSK-equivalent, Confluent, Event Hubs Kafka endpoint) or Pub/Sub; rewrite producers/consumers against the Kafka API to avoid the next lock-in. | **Azure:** Event Hubs (Kafka API)<br>**GCP:** Pub/Sub / Managed Kafka<br>**Neutral:** Apache Kafka / Redpanda | 1-3 months |
| **Cognito**<br>Security<br>$7.3k/mo in scope | User pools hold customer identities; password hashes cannot be exported, and triggers / hosted UI are AWS-specific. Customers notice an identity migration. | Move to Entra External ID, Identity Platform, Auth0 or Keycloak using lazy (on-login) migration; plan MFA re-enrolment and app token changes. | **Azure:** Entra External ID<br>**GCP:** Identity Platform<br>**Neutral:** Auth0 / Keycloak / Okta CIC | 2-4 months (customer-facing) |
| **Step Functions**<br>Integration<br>$6.4k/mo in scope | Workflows are written in Amazon States Language with direct service integrations to 200+ AWS APIs; nothing executes ASL outside AWS. | Rewrite workflows in Logic Apps / Durable Functions, Google Workflows, or Temporal (cloud-neutral and code-first). Redesign each service integration step. | **Azure:** Logic Apps / Durable Functions<br>**GCP:** Workflows<br>**Neutral:** Temporal / Argo Workflows | 1-3 months |
| **QuickSight**<br>Analytics<br>$5.6k/mo in scope | Proprietary BI tool; dashboards, SPICE datasets and embedding cannot be exported to other BI products. | Rebuild priority dashboards in Power BI, Looker or Superset; retire the long tail. Treat as a business change, not a technical migration. | **Azure:** Power BI<br>**GCP:** Looker<br>**Neutral:** Apache Superset / Tableau | 1-3 months |
| **AppSync**<br>Integration<br>$3.9k/mo in scope | GraphQL is standard but AppSync resolvers (VTL / APPSYNC_JS), data-source bindings and subscriptions are proprietary. | Re-implement with Apollo Server / Hasura / Azure API Management GraphQL; rewrite resolvers. | **Azure:** API Management (GraphQL) / Hasura<br>**GCP:** Apollo on Cloud Run<br>**Neutral:** Apollo / Hasura | 1-3 months |
| **Lex**<br>AI/ML<br>$2.8k/mo in scope | Proprietary bot definitions, intents and Connect integration. | Rebuild bots in Copilot Studio / Dialogflow CX or an LLM-based assistant. | **Azure:** Copilot Studio<br>**GCP:** Dialogflow CX<br>**Neutral:** Rasa / LLM agents | 1-3 months |
| **Pinpoint**<br>Business apps<br>$2.6k/mo in scope | Proprietary campaign, journey and segment definitions; sender IDs and registered numbers are tied to AWS. | Move to Braze / Twilio / Azure Communication Services; re-register sender IDs and short codes. | **Azure:** Azure Communication Services<br>**GCP:** Firebase Cloud Messaging + partner<br>**Neutral:** Braze / Twilio | 1-3 months |
| **Timestream**<br>Database<br>$1.8k/mo in scope | Proprietary time-series engine and SQL dialect (Timestream for InfluxDB is the portable variant). | Move to InfluxDB, TimescaleDB, Azure Data Explorer or Bigtable; rewrite queries and ingestion. | **Azure:** Azure Data Explorer<br>**GCP:** Bigtable / BigQuery<br>**Neutral:** InfluxDB / TimescaleDB | 1-3 months |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **EventBridge**<br>Integration<br>$1.4k/mo in scope | Proprietary event bus whose value is AWS-native event sources, rules, Pipes and the Scheduler. Event patterns and targets are AWS ARNs. | Re-implement routing on Event Grid / Eventarc or Kafka; replace AWS-native sources with the target's audit/event feeds; move schedules to the target scheduler. | **Azure:** Event Grid<br>**GCP:** Eventarc / Cloud Scheduler<br>**Neutral:** Kafka / Argo Events | 1-2 months |
| **Amplify**<br>DevOps<br>$1.2k/mo in scope | Opinionated framework that generates Cognito, AppSync, DynamoDB and Lambda backends; front-end hosting is portable, the generated backend is not. | Host front-ends on Static Web Apps / Firebase Hosting / Vercel; re-platform the backend along with Cognito, AppSync and DynamoDB. | **Azure:** Static Web Apps<br>**GCP:** Firebase Hosting<br>**Neutral:** Vercel / Netlify | 1-3 months |
| **Bedrock AgentCore**<br>AI/ML<br>$1.1k/mo in scope | Agent runtime, memory, gateway, identity and observability services with AWS-specific APIs. Agent code written with open frameworks is portable; the hosting, memory stores and tool gateway are not. | Keep agent logic in an open framework (LangGraph, Strands, Claude Agent SDK); re-host on Azure AI Foundry Agent Service, Vertex AI Agent Engine or containers; migrate memory to a vector/relational store you control. | **Azure:** Azure AI Foundry Agent Service<br>**GCP:** Vertex AI Agent Engine<br>**Neutral:** Containers + open agent frameworks | 1-2 months |
| **Lake Formation**<br>Analytics<br>$604/mo in scope | Proprietary fine-grained permission layer over the Glue catalog; access policies must be re-modelled. | Re-implement as Unity Catalog, Microsoft Purview / OneLake security, or Dataplex policies. | **Azure:** Microsoft Purview / OneLake security<br>**GCP:** Dataplex / BigLake<br>**Neutral:** Unity Catalog / Apache Ranger | 1-2 months |
| **Data Pipeline**<br>Analytics<br>$291/mo in scope | Legacy proprietary orchestration (in maintenance mode). | Rewrite as Airflow DAGs or Data Factory pipelines. | **Azure:** Data Factory<br>**GCP:** Cloud Composer<br>**Neutral:** Apache Airflow | 1-2 months |

**Tier 3 · Moderate** **10 service(s)**

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **SageMaker**<br>AI/ML<br>$21k/mo in scope | Models (PyTorch, XGBoost, Hugging Face) and containers are portable, but pipelines, feature store, endpoints and Studio domains are SageMaker-specific. | Move training/inference to Azure ML / Vertex AI or Kubeflow/MLflow; rebuild pipelines and feature store. | **Azure:** Azure Machine Learning<br>**GCP:** Vertex AI<br>**Neutral:** Kubeflow / MLflow / KServe | 1-3 months |
| **Bedrock**<br>AI/ML<br>$9.3k/mo in scope | Foundation models (Anthropic Claude, Llama, Mistral) are available on other clouds and direct from providers, but Bedrock Agents, Knowledge Bases, Guardrails and Flows are proprietary orchestration. | Call the same models via Vertex AI / Azure AI Foundry or the provider's own API; re-implement agents and RAG pipelines with a portable framework; re-run evaluations. | **Azure:** Azure AI Foundry<br>**GCP:** Vertex AI<br>**Neutral:** Provider APIs (Anthropic | 3-8 weeks |
| **API Gateway**<br>Integration<br>$8.6k/mo in scope | OpenAPI definitions export, but Lambda/VTL integrations, authorizers, usage plans and WebSocket routes are AWS-specific. | Export OpenAPI, re-implement on Azure API Management, Apigee or Kong; rewrite authorizers and mapping templates. | **Azure:** API Management<br>**GCP:** Apigee / API Gateway<br>**Neutral:** Kong / Envoy Gateway | 3-8 weeks |
| **Athena**<br>Analytics<br>$6.2k/mo in scope | Trino/Presto-based SQL, so queries are largely portable, but it is bound to the Glue Data Catalog, S3 locations and Athena workgroups. | Move tables to Iceberg/Delta, query with Trino, BigQuery or Databricks SQL. | **Azure:** Fabric / Synapse serverless SQL<br>**GCP:** BigQuery<br>**Neutral:** Trino / Starburst | 3-8 weeks |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **Neptune**<br>Database<br>$6.1k/mo in scope | Graph database using open query languages (Gremlin, openCypher, SPARQL), but loaders, IAM auth and Neptune-specific extensions need rework. | Move to Neo4j, Cosmos DB for Gremlin or JanusGraph; re-validate query behaviour. | **Azure:** Cosmos DB for Apache Gremlin<br>**GCP:** Neo4j AuraDB / Spanner Graph<br>**Neutral:** Neo4j / JanusGraph | 1-3 months |
| **Transcribe**<br>AI/ML<br>$3.0k/mo in scope | Proprietary speech API; custom vocabularies need recreating. | Swap to Azure AI Speech / Speech-to-Text or Whisper. | **Azure:** Azure AI Speech<br>**GCP:** Speech-to-Text<br>**Neutral:** Whisper | 2-4 weeks |
| **SQS**<br>Integration<br>$2.8k/mo in scope | Simple semantics, but a proprietary API; every producer and consumer needs a code change. | Move to Service Bus / Pub/Sub / RabbitMQ behind a small messaging abstraction; drain queues at cutover. | **Azure:** Service Bus / Storage Queues<br>**GCP:** Pub/Sub / Cloud Tasks<br>**Neutral:** RabbitMQ / NATS | 2-6 weeks |
| **Textract**<br>AI/ML<br>$2.6k/mo in scope | Proprietary document-extraction API and output schema. | Move to Azure Document Intelligence / Document AI; re-map output parsing and re-validate accuracy. | **Azure:** Document Intelligence<br>**GCP:** Document AI<br>**Neutral:** Open-source OCR + LLM extraction | 1-2 months |
| **Firehose**<br>Analytics<br>$2.4k/mo in scope | Managed delivery pipeline with AWS-only sources and destinations. | Replace with Kafka Connect, Event Hubs Capture, or Dataflow templates. | **Azure:** Event Hubs Capture / Stream Analytics<br>**GCP:** Dataflow / Pub/Sub subscriptions<br>**Neutral:** Kafka Connect / Vector | 3-6 weeks |
| **SNS**<br>Integration<br>$1.7k/mo in scope | Proprietary pub/sub and SMS/mobile push API, usually fanned out to SQS and Lambda. | Move topics to Event Grid / Service Bus topics / Pub/Sub; move SMS to a CPaaS provider (Twilio, Azure Communication Services). | **Azure:** Event Grid / Service Bus topics<br>**GCP:** Pub/Sub<br>**Neutral:** NATS / Kafka + Twilio | 2-6 weeks |

**Tier 2 · Low** **17 service(s)**

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **RDS / Aurora**<br>Database<br>$64k/mo in scope | PostgreSQL, MySQL, MariaDB, SQL Server and Oracle engines, plus Aurora (PostgreSQL/MySQL compatible). Schemas and SQL move; Aurora-only features (Global Database, Babelfish, Aurora Serverless scaling, IAM DB auth) and Oracle/SQL Server licensing need attention. | Migrate with native replication or DMS-equivalents (Azure DMS, Database Migration Service) into Azure Database / Cloud SQL / AlloyDB. Cut over per database with a short read-only window. | **Azure:** Azure Database for PostgreSQL/MySQL / Azure SQL<br>**GCP:** Cloud SQL / AlloyDB<br>**Neutral:** Self-managed PostgreSQL (CloudNativePG) | 2-8 weeks per database |
| **S3**<br>Storage<br>$37k/mo in scope | The S3 API is a de-facto standard implemented by most object stores, but bucket policies, event notifications, lifecycle rules, Object Lock and presigned URL patterns need translating, and large data sets incur egress cost and transfer time. | Copy with rclone / Storage Transfer Service / AzCopy; use S3-compatible endpoints where code depends on the S3 SDK; recreate lifecycle, retention and event rules. | **Azure:** Blob Storage<br>**GCP:** Cloud Storage<br>**Neutral:** MinIO / Ceph RGW | 1-8 weeks (data-volume bound) |
| **Direct Connect**<br>Networking<br>$10k/mo in scope | Physical circuits; lead times for new carrier connections dominate. | Order ExpressRoute / Cloud Interconnect early (8-16 week lead time is common). | **Azure:** ExpressRoute<br>**GCP:** Cloud Interconnect<br>**Neutral:** Carrier MPLS | 2-4 months (carrier lead time) |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **EMR**<br>Analytics<br>$9.9k/mo in scope | Open-source Spark, Hive, Presto, Flink. EMRFS, bootstrap actions and step orchestration need adjusting. | Run the same jobs on Databricks, Dataproc or HDInsight; swap s3:// paths and IAM roles. | **Azure:** Databricks / HDInsight<br>**GCP:** Dataproc<br>**Neutral:** Spark on Kubernetes | 1-2 months |
| **ECS**<br>Compute<br>$9.6k/mo in scope | Containers are portable but ECS task definitions, service discovery and deployment wiring are AWS-specific and must be rewritten. | Convert task definitions to Kubernetes manifests or Azure Container Apps / Cloud Run definitions. Images move unchanged. | **Azure:** Container Apps / AKS<br>**GCP:** Cloud Run / GKE<br>**Neutral:** Kubernetes / Nomad | 2-6 weeks per service group |
| **CloudWatch**<br>Management<br>$7.5k/mo in scope | Metrics, logs, alarms and dashboards are re-creatable, but Embedded Metric Format, Logs Insights queries and alarm-driven automation are AWS-specific. | Standardise on OpenTelemetry and ship to Azure Monitor / Cloud Monitoring / Grafana stack; rebuild alarms and dashboards from code. | **Azure:** Azure Monitor<br>**GCP:** Cloud Monitoring / Logging<br>**Neutral:** Grafana / Prometheus / Loki / Datadog | 2-4 weeks |
| **CloudFront**<br>Networking<br>$6.9k/mo in scope | A CDN like any other, but Lambda@Edge / CloudFront Functions and signed-URL schemes are proprietary. | Move to Front Door, Cloud CDN, Cloudflare or Akamai; rewrite edge functions. | **Azure:** Front Door<br>**GCP:** Cloud CDN<br>**Neutral:** Cloudflare / Akamai / Fastly | 1-3 weeks |
| **Network Firewall**<br>Networking<br>$4.7k/mo in scope | Suricata-compatible rules; policy structure is AWS-specific. | Translate rules to Azure Firewall / Cloud NGFW or a vendor NGFW. | **Azure:** Azure Firewall<br>**GCP:** Cloud NGFW<br>**Neutral:** Palo Alto / Fortinet | 2-4 weeks |
| **Managed Flink**<br>Analytics<br>$3.2k/mo in scope | Apache Flink applications are portable; legacy Kinesis SQL applications are not. | Run Flink jobs on Confluent, Ververica, or Flink on Kubernetes; rewrite any legacy SQL apps. | **Azure:** Flink on HDInsight on AKS / Stream Analytics<br>**GCP:** Dataflow / Flink on GKE<br>**Neutral:** Apache Flink | 1-2 months |
| **Backup**<br>Storage<br>$3.0k/mo in scope | Backup policies and vaults are AWS-specific; the protected data moves with each workload. Retained backups must be kept (or restored and re-protected) to meet retention obligations. | Recreate policies in Azure Backup / Backup and DR; keep the AWS vault read-only until retention expires. | **Azure:** Azure Backup<br>**GCP:** Backup and DR<br>**Neutral:** Veeam / Commvault | Retention-bound tail |
| **KMS**<br>Security<br>$2.1k/mo in scope | Keys cannot be exported from KMS. Data encrypted under KMS must be decrypted and re-encrypted on the target; envelope-encryption code calls the KMS API. | Create keys in Key Vault / Cloud KMS (or an external HSM / BYOK), re-encrypt data during migration, swap the SDK calls. | **Azure:** Key Vault / Managed HSM<br>**GCP:** Cloud KMS / Cloud HSM<br>**Neutral:** HashiCorp Vault / Thales HSM | Moves with the workload |
| **Directory Service**<br>Security<br>$2.1k/mo in scope | Managed Microsoft AD - standard Active Directory. | Extend trusts to Entra Domain Services / Managed AD on the target and migrate members. | **Azure:** Entra Domain Services<br>**GCP:** Managed Service for Microsoft AD<br>**Neutral:** Active Directory | 1-2 months |
| **ACM / Private CA**<br>Security<br>$1.4k/mo in scope | Public certificates are free to reissue elsewhere; a private CA hierarchy must be re-established and trusted by every client. | Reissue public certs on the target; plan a private CA trust rollout (or adopt a cloud-neutral CA such as Vault PKI). | **Azure:** Key Vault certificates<br>**GCP:** Certificate Authority Service<br>**Neutral:** Vault PKI / EJBCA | 2-6 weeks |
| **WAF**<br>Security<br>$1.4k/mo in scope | Rule concepts are common; rule syntax and managed rule groups are AWS-specific. | Translate rules to Azure WAF / Cloud Armor / Cloudflare WAF; re-tune in detection mode before enforcing. | **Azure:** Azure WAF<br>**GCP:** Cloud Armor<br>**Neutral:** Cloudflare / Akamai WAF | 2-4 weeks |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **Global Accelerator**<br>Networking<br>$691/mo in scope | Anycast IP front door; static IPs cannot be moved. | Replace with Front Door / global load balancer; plan for client IP allow-list changes. | **Azure:** Front Door / cross-region Load Balancer<br>**GCP:** Global external Load Balancer<br>**Neutral:** Cloudflare Spectrum | 2-4 weeks |
| **Secrets Manager**<br>Security<br>$589/mo in scope | Simple secret store, but rotation Lambdas and SDK calls are AWS-specific. | Move secrets to Key Vault / Secret Manager / Vault; rewrite rotation; prefer External Secrets Operator for Kubernetes workloads. | **Azure:** Key Vault<br>**GCP:** Secret Manager<br>**Neutral:** HashiCorp Vault | 1-2 weeks |
| **Batch**<br>Compute<br>$0/mo in scope | Container jobs on EC2/Fargate; job definitions and queues are AWS-specific. | Move to Azure Batch, Google Batch, or Kubernetes Jobs / Argo Workflows. | **Azure:** Azure Batch<br>**GCP:** Google Batch<br>**Neutral:** Kubernetes Jobs / Slurm | 2-4 weeks |

**Tier 1 · Portable** 12 service(s)

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **EC2**<br>Compute<br>$106k/mo in scope | Virtual machines running standard operating systems. Images can be exported or rebuilt from configuration management. | Rehost VMs (Azure Migrate, Google Migrate to VMs, or rebuild from Packer/Ansible images). Re-baseline instance sizes on the target. | **Azure:** Virtual Machines<br>**GCP:** Compute Engine<br>**Neutral:** VMware / OpenStack / bare metal | 1-4 weeks per application |
| **EC2 Other (EBS, NAT, transfer)**<br>Compute<br>$33k/mo in scope | Block storage, snapshots, NAT gateways, elastic IPs and data transfer. Commodity primitives on every cloud. | Recreate volumes and network egress on the target; move data with snapshots or block replication. Budget for one-off egress charges. | **Azure:** Managed Disks / NAT Gateway<br>**GCP:** Persistent Disk / Cloud NAT<br>**Neutral:** SAN / Ceph | Moves with the workload |
| **VPC**<br>Networking<br>$11k/mo in scope | Standard network constructs (subnets, routing, VPN, IPAM); only the console and APIs differ. | Design the target landing-zone network; reuse the IP plan where possible. | **Azure:** Virtual Network<br>**GCP:** VPC<br>**Neutral:** Data-centre networking | Part of landing zone |
| **MSK (Kafka)**<br>Analytics<br>$7.1k/mo in scope | Standard Apache Kafka. | Mirror topics with MirrorMaker 2 / Cluster Linking and cut consumers over. | **Azure:** Event Hubs (Kafka) / Confluent<br>**GCP:** Managed Service for Apache Kafka<br>**Neutral:** Apache Kafka / Redpanda | 2-6 weeks |
| **FSx**<br>Storage<br>$5.0k/mo in scope | Managed Windows File Server, Lustre, ONTAP or OpenZFS - all standard file systems. | Replicate to the equivalent managed file system (ONTAP SnapMirror where applicable). | **Azure:** Azure Files / NetApp Files<br>**GCP:** Filestore / NetApp Volumes<br>**Neutral:** NetApp / Windows File Server | 2-4 weeks |
| **OpenSearch**<br>Analytics<br>$4.2k/mo in scope | Open-source OpenSearch / Elasticsearch APIs (Serverless collections have a reduced API). | Snapshot/restore or reindex into Elastic Cloud, OpenSearch on Kubernetes or Azure AI Search where suitable. | **Azure:** Elastic Cloud on Azure<br>**GCP:** Elastic Cloud on GCP<br>**Neutral:** OpenSearch | 2-4 weeks |
| **EKS**<br>Compute<br>$4.1k/mo in scope | Upstream Kubernetes. Manifests and Helm charts run on any conformant cluster; AWS-specific add-ons (IRSA, ALB controller, EBS CSI) are small and swappable. | Stand up AKS/GKE/on-prem Kubernetes, swap cloud add-ons (identity, ingress, storage classes), redeploy via GitOps. | **Azure:** AKS<br>**GCP:** GKE<br>**Neutral:** OpenShift / Rancher / upstream Kubernetes | 2-6 weeks per cluster |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **ELB**<br>Networking<br>$3.8k/mo in scope | Standard L4/L7 load balancing. | Recreate listeners and rules on Application Gateway / Cloud Load Balancing / Envoy. | **Azure:** Load Balancer / Application Gateway<br>**GCP:** Cloud Load Balancing<br>**Neutral:** HAProxy / F5 / Envoy | Moves with the workload |
| **ElastiCache**<br>Database<br>$3.7k/mo in scope | Managed Redis / Valkey / Memcached - open engines. | Provision the target cache and warm it; caches rarely need data migration. | **Azure:** Azure Cache for Redis<br>**GCP:** Memorystore<br>**Neutral:** Redis / Valkey | Days |
| **MWAA (Airflow)**<br>Analytics<br>$1.9k/mo in scope | Apache Airflow; DAGs are portable apart from AWS operators. | Move DAGs to Cloud Composer, Astronomer or Airflow on Kubernetes; swap AWS operators. | **Azure:** Data Factory Workflow Orchestration Manager / Astronomer<br>**GCP:** Cloud Composer<br>**Neutral:** Apache Airflow | 2-4 weeks |
| **Amazon MQ**<br>Integration<br>$1.3k/mo in scope | Managed ActiveMQ / RabbitMQ - standard protocols (AMQP, MQTT, STOMP, JMS). | Shovel/federate queues to the target broker and cut clients over. | **Azure:** Service Bus (AMQP) / RabbitMQ on AKS<br>**GCP:** RabbitMQ on GKE<br>**Neutral:** RabbitMQ / ActiveMQ | 1-3 weeks |
| **Route 53**<br>Networking<br>$891/mo in scope | DNS zones export as standard zone files. Alias records, health-check routing and Resolver rules need translating. | Export zones, recreate in Azure DNS / Cloud DNS / NS1, then lower TTLs and switch NS records. | **Azure:** Azure DNS<br>**GCP:** Cloud DNS<br>**Neutral:** NS1 / Cloudflare DNS | 1-2 weeks |

### Unclassified services in this estate

Scored as Moderate (tier 3) until added to the catalog.

- AWS Transform — $437/month

# B Appendix: methodology

## Data sources

Spend comes from AWS Cost Explorer in the management (payer) account: amortised cost per account per service per month, excluding tax, credits, refunds and support. Marketplace purchases are grouped as one third-party line. Accounts and OU structure come from AWS Organizations. The optional deep scan counts resources (Lambda functions, DynamoDB tables, state machines, stacks and more) in each member account through a read-only role. All API calls are read-only.

## Scoring

Each service's tier maps to a score: Portable 10, Low 30, Moderate 50, High 75, Critical 95. An account's **inherent risk** is the spend-weighted average of its services' scores. Its **weight** is its share of in-scope spend, and its **contribution** is inherent risk × weight. The **OU score** is the sum of contributions — the spend-weighted risk of the whole estate — so a small account full of proprietary services scores high on its own yet moves the OU score very little.

Ratings: Low < 25 ≤ Moderate < 40 ≤ Elevated < 55 ≤ High < 70 ≤ Critical.

## Estimate

Effort is built bottom-up: each material service in each account contributes the catalog's engineer-weeks for one workload, scaled by the square root of its spend relative to US$1,000/month (and by its resource count where the deep scan ran). Account overhead, a target landing zone, programme management and contingency are added, and the total is adjusted for delivery maturity from the questionnaire. Duration is the longer of effort ÷ available capacity and the critical path of the hardest tier present.

## Limitations

- Spend is a proxy for size; a cheap service can still be hard to move (the deep scan's resource counts partly correct for this).
- RDS spend covers all engines; Aurora-specific features and commercial engines (Oracle, SQL Server) appear in the deep scan's engine breakdown.
- Contractual exit costs (EDP commitments, Savings Plans, Reserved Instances) and target-cloud run costs are outside the estimate.
- The estimate is an order-of-magnitude planning figure for an executive decision, not a delivery quote.